



RENEWABLE ENERGY TRENDS IN 2019

AS THE YEAR COMES TO A CLOSE, WE HIGHLIGHT SOME OF THE PROMISING RENEWABLE ENERGY TRENDS THAT MATURED IN 2019

Dhriti Datta | dhriti@digit.in

A decade ago, it was considered almost atrocious to intimate that renewable energy could ever overtake fuel and the other conventional energy sources. The global demand for coal was on pace to ramp up by a staggering 53 per cent within two decades and wind and solar energy comprised a mere one per cent of the United States' energy sources. Fast-forward to April 2019, renewable energy sources are now mainstream options in the power sector in nearly every nook and cranny of the world. According to BloombergNEF forecasts, solar and wind energy will power half the world by 2050. Predictions indicate that renewable power plants will outnumber their non-renewable counterparts by vast margins in 2025.

Last year, the CDP enlisted over 100 cities across the globe that reported 70 per cent of their electricity is derived from renewables, including Auckland, Seattle and more. Furthermore, 40 cities among those have currently attained 100 per cent electricity procured from renewable sources. These renewable sources (especially wind and solar),

however, aren't proliferating as much due to moral do-gooders, but because of how immensely profitable they are to the power businesses across most of the world. A power company that was once relying on hefty subsidies and being propped up by government mandates suddenly has the ability to stand on their own feet. Plummeting costs of renewable sources and wide scale opportunities derived from growing technological prowess have accelerated the adoption of sustainable energy sources as well. With countries and companies embracing sustainable and renewable energy, we are now witnessing next-gen, renewable energy trends, some of which are heavily tech-aided, in 2019. Let's delve into these.

ENERGY STORAGE

Possibly one of the most integral energy trends today that is gaining popularity is energy storage. It plays a crucial role in balancing the demand and supply of power and also has the capability to tackle intermittency issues of renewable energy sources. Coupling a renewable energy source with a storage system guarantees a steady supply of power,

even when weather conditions are less than ideal for energy generation.

Batteries are the customary storage devices used in renewable energy solutions and their adaptation is escalating on a residential and grid-wide scale level. The storage options are expected to become more viable and affordable in a few years. According to projections, storage is touted to be the core component that will make renewable sources more accessible and prevalent as the prices of energy storage drop consistently. This will essentially erode the advantages of traditional energy sources.

DEPLOYMENT OF MICROGRIDS AND AI

Microgrids have astronomically surged in popularity since they provide energy independence, efficiency and protection during calamities. They are local energy grids that can work autonomously or when connected to larger conventional grids. The power of AI and ML has been integrated into microgrid controllers. This facilitates constant adaptation and improvement of operation as per the requirement calculated by the AI. The process of deploying these microgrids has